

AfyaIntel Hub: A Generative Health Intelligence and Clinical NLP Framework for Predictive Supply Chains and Offline Decision Support in Low-Resource Settings

Prepared by

Majaliwa Tani Mabirika

Independent Researcher and Innovator | B.Sc. with Education in Mathematics and Informatics Studies |
University of Dar es Salaam, Mkwawa University College of Education (MUCE)

Table FM1. Author and Manuscript Identification

Item	Details
Research artifact	AfyaIntel Hub
Artifact type	Offline-first generative health intelligence, clinical NLP, predictive supply-chain and interoperability framework
Research paradigm	Design Science Research Methodology (DSRM)
Geographic focus	Low-resource primary healthcare settings in Tanzania
Clinical boundary	Decision support only. The artifact does not replace licensed healthcare professionals, clinical judgment, regulatory authorization, or official national electronic medical records.

Submission status: Final research manuscript prepared for institutional review, health informatics peer review, research partnership screening, grant assessment, and controlled pilot authorization.

Declaration

I, Majaliwa Tani Mabirika, declare that this research manuscript and the proposed AfyaIntel Hub software artifact are original intellectual and technical work prepared for academic, innovation, and institutional health-sector review. The manuscript presents a Design Science Research contribution centered on a health informatics artifact for low-resource clinical environments. It does not present the artifact as an autonomous diagnostic engine and does not claim clinical efficacy without ethical approval, formal validation, and qualified clinical review.

All patient-facing, clinician-facing, and medicine-supply functions described in this manuscript are designed as decision-support, coordination, analytics, and learning-health-system services. Any use of real patient data, facility records, prescription data, or national health information systems shall require authorization from relevant institutional authorities, ethical review bodies, data protection offices, and clinical governance committees.

Signature: _____

Name: Majaliwa Tani Mabirika

Date: _____

Certification

This page is reserved for formal certification by academic supervisors, university reviewers, institutional health partners, and authorized pilot-site representatives. Certification confirms that the manuscript has been reviewed as a health informatics Design Science Research artifact and that any pilot implementation will be subject to ethical clearance, data protection registration where applicable, facility authorization, and clinical safety oversight.

Table FM2. Certification and Review Signature Block

Reviewer category	Name and title	Institution	Signature and date
Academic supervisor			
University reviewer			
Clinical partner			
Digital health or data governance partner			
Pilot facility representative			

Dedication

This work is dedicated to Tanzanian medical professionals, community health workers, rural clinical officers, nurses, pharmacists, health information officers, and facility managers who continue to deliver essential healthcare in low-resource environments despite limited personnel, intermittent connectivity, constrained medicine availability, and operational pressure.

The AfyaIntel Hub concept honors the practical intelligence of frontline health workers by designing technology that supports their workflow rather than replacing their judgment.

Acknowledgements

The author acknowledges the contribution of Tanzania's broader digital health ecosystem, including the Ministry of Health, regional and district health management structures, local health facility managers, academic mentors at Mkwawa University College of Education, and the open-source Swahili and African-language NLP research communities whose work makes localized health informatics innovation possible.

Special appreciation is extended to the developers, health informatics practitioners, clinical officers, nurses, pharmacists, public-health researchers, and open-source communities working on interoperability, low-resource NLP, offline-first systems, and responsible medical AI. Their collective knowledge informs the technical and ethical direction of AfyaIntel Hub.

Abstract

Primary healthcare facilities in low-resource settings often operate under structural constraints that include paper-based records, intermittent electricity, unreliable internet connectivity, fragmented health information systems, limited clinical decision-support tools, and delayed pharmaceutical stock reporting. These constraints reduce the speed and reliability of symptom triage, disease intelligence, and medicine supply planning. This paper presents AfyaIntel Hub, a Design Science Research artifact that integrates an offline-first clinical workflow, Swahili and Swanglish clinical natural language processing, predictive pharmaceutical forecasting, and HL7 FHIR-based interoperability for low-resource primary care settings in Tanzania. The artifact is designed around a local SQLite data store, encrypted synchronization queues, role-based access control, a bilingual clinical NLP gateway based on SwahBERT or AfroXLMR adaptation, and a forecasting engine using XGBoost and Prophet to estimate drug consumption trends and potential stock-out windows. The system maps local observations, encounters, medication stock records, and facility indicators into FHIR-compatible resources for safe exchange with national systems such as GoT-HOMIS and DHIS2. Evaluation is structured through DSRM-aligned artifact validation using NLP precision, recall and F1-score; forecasting MAE, RMSE and WAPE; offline synchronization reliability; System Usability Scale questionnaires; and human-in-the-loop clinical safety review. AfyaIntel Hub contributes a localized health informatics framework for connecting point-of-care symptom intelligence, clinical support, disease surveillance, and supply-chain planning without replacing licensed clinical decision-making. The paper concludes that responsible, offline-capable, language-aware, and interoperable medical AI can strengthen health system resilience when governed through privacy, clinical accountability, and institutional partnership.

Keywords: Clinical NLP; Pharmaceutical Forecasting; Low-Resource Healthcare; HL7 FHIR; Data Sovereignty; Swahili-BERT; Offline-First Systems; AfroXLMR; GoT-HOMIS; DHIS2; Design Science Research; Tanzania Digital Health.

Table of Contents

Front Matter

Chapter 1: Introduction

Chapter 2: Literature Review and Gap Analysis

Chapter 3: Research Methodology (DSRM Framework)

Chapter 4: Conceptual Framework and System Design

Chapter 5: Database Design, AI/ML Workflow, and Mathematical Framework

Chapter 6: Multi-Layer Operational Framework and Cost Model

Chapter 7: Security, Privacy, and Human-in-the-Loop Governance

Chapter 8: Prototyping, Pilot Evaluation, and System Performance

Chapter 9: Scientific Contribution and Practical Horizons

References

Appendices

Chapter 1: Introduction

1.1 Background of the Study

Tanzania's primary healthcare delivery environment depends heavily on dispensaries, health centers, district hospitals, regional hospitals, community health workers, nurses, clinical officers, pharmacists, laboratory staff, and health information officers. At the lower levels of care, point-of-care documentation may still rely on paper registers, locally maintained spreadsheets, partially digitized systems, and delayed district-level reporting. These workflows create a gap between what happens during patient encounters and what becomes available for regional disease intelligence, clinical quality improvement, and pharmaceutical supply planning.

Digital health systems have become central to global health-system strengthening. The World Health Organization emphasizes that digital interventions should be evaluated for effectiveness, feasibility, acceptability, resource use, equity, and implementation context. Tanzania's National Digital Health Strategy 2019-2024 similarly recognizes the need to strengthen digital health infrastructure and health information exchange in support of universal health coverage. AfyaIntel Hub is positioned within this national and global transition toward interoperable, evidence-informed, and accountable digital health systems.

The technical purpose of AfyaIntel Hub is to create a local intelligence layer that can operate inside low-resource primary care workflows. The system captures Swahili, English, and Swanglish symptom narratives; extracts medically relevant concepts; supports human-reviewed triage prompts; verifies local medicine availability; forecasts stock-out risk; and synchronizes de-identified or authorized data with regional dashboards when connectivity becomes available.

1.2 Problem Statement

Frontline health posts and lower-level facilities face structural vulnerabilities that reduce the timeliness and quality of care coordination. These include delayed paper ledger updates, limited localized digital triage support, language mismatch between patient descriptions and structured medical vocabularies, incomplete historical stock records, and chronic power or network outages. Existing systems may support clinical records, public health reporting, or inventory management separately, but they do not always provide a unified workflow that connects symptom intelligence, clinical review, and medicine demand forecasting at the point of care.

This fragmentation produces four practical consequences. First, clinical workers may spend additional time converting patient narratives into structured records. Second, medicine consumption patterns may be reported after shortages have already affected service delivery. Third, district and regional managers may receive delayed or incomplete signals about emerging disease patterns. Fourth, digital tools that require continuous internet connectivity may fail precisely in the environments that need decision support most.

AfyaIntel Hub responds to this problem by proposing a secure, offline-first, language-aware, human-in-the-loop health intelligence artifact that links point-of-care symptom entry with clinical decision support, medicine stock awareness, predictive forecasting, and interoperable public-health reporting.

1.3 General Objective

The general objective of this study is to design, develop, and evaluate AfyaIntel Hub as a secure, offline-first generative health intelligence framework that improves localized diagnostic triage entry,

supports human-reviewed clinical decision-making, and stabilizes pharmaceutical supply networks in low-resource primary healthcare settings in Tanzania.

1.4 Specific Objectives

Table 1. Specific Objectives and Technical Deliverables

No.	Objective	Technical deliverable
1	Design a modular software architecture supporting local SQLite databases, message-queuing for synchronization, and interactive regional dashboards.	Offline client application, API gateway, sync service, dashboard service, and facility-level data cache.
2	Develop a bilingual clinical NLP gateway optimized to extract symptoms and medical intents from informal Swahili and Swanglish clinical notes.	NLP pipeline using SwahBERT/AfroXLMR adaptation, symptom dictionary, red-flag detector, and intent classifier.
3	Develop a predictive supply-chain engine that calculates local consumption trends and generates early warning alerts for impending medicine stock-outs.	XGBoost and Prophet forecasting service, stock-out window estimator, and alert prioritization logic.
4	Formulate an interoperable data-mapping layer using HL7 FHIR protocols for safe communication with national platforms such as GoT-HOMIS and DHIS2.	FHIR R4 mapping templates for Patient, Encounter, Observation, MedicationRequest, MedicationStatement, Organization, Location, and aggregate indicator bundles.
5	Establish data-governance protocols that maintain clinical accountability and align with the Tanzanian Personal Data Protection Act, 2022.	Consent model, data minimization rules, anonymization logic, RBAC, audit logs, and data protection assessment checklist.

1.5 Research Questions

Table 2. Research Questions and Evaluation Linkage

Research question	Artifact component	Evaluation indicator
How accurately can localized clinical NLP extract symptoms, urgency indicators, and intents	Clinical NLP Gateway	Precision, recall, F1-score, confusion matrix by symptom category and dialect subset.

Research question	Artifact component	Evaluation indicator
from Swahili and Swanglish text?		
How can forecasting models handle fragmented medicine stock histories and seasonal disease patterns in lower-level facilities?	Predictive Supply Chain Engine	MAE, RMSE, WAPE, stock-out detection lead time, missing-data robustness.
How can offline-first architecture preserve clinical workflow continuity during power and network outages?	SQLite cache and sync queue	Sync success rate, queue recovery time, local load time, data conflict resolution accuracy.
How can AfyaIntel Hub interoperate with national health information systems without becoming a replacement for official EMR platforms?	FHIR Interoperability Mapper	FHIR validation success rate, field completeness, DHIS2 aggregate mapping coverage.
What governance mechanisms are required to keep AI outputs under human clinical control?	Human-in-the-loop governance	Override rate, clinical review completion rate, auditability, role-based access compliance.

1.6 Scope of the Study

The study focuses on low-resource primary care facilities, including dispensaries and health centers that may have limited connectivity, limited power reliability, and mixed paper-digital workflows. The artifact is designed as an intelligence layer, not as a replacement for official electronic medical records, national reporting systems, licensed diagnostic authority, or regulatory clinical protocols.

The technical scope covers offline clinical note capture, localized NLP parsing, human-reviewed triage support, medicine inventory checks, forecasting alerts, data synchronization, FHIR-compatible mapping, privacy controls, and dashboard analytics. The clinical scope is limited to decision support, education prompts, red-flag identification, and stock-planning intelligence. Any use for direct patient management must occur under qualified clinical supervision and approved facility procedures.

1.7 Significance of the Study

Table 3. Multi-Stakeholder Value Matrix

Stakeholder	Operational challenge	AfyaIntel Hub contribution
Clinical officers and nurses	High workload, limited digital decision support, and informal patient descriptions.	Structured symptom capture, urgency prompts, guideline references, and offline access.
Facility pharmacists and	Stock ledgers may be delayed,	Stock balance visibility,

Stakeholder	Operational challenge	AfyaIntel Hub contribution
storekeepers	incomplete, or disconnected from disease trends.	consumption forecasting, and early warning alerts.
District and regional health managers	Delayed visibility into disease clusters and facility supply pressure.	Synchronized dashboards for aggregate disease and stock indicators.
Health ministries and public health programs	Need interoperable, accountable, and privacy-preserving health data streams.	FHIR-compatible mapping and governance-oriented analytics.
Health informatics researchers	Limited African-language clinical NLP and offline-first design literature.	A localized DSRM artifact for Swahili/Swanglish clinical text and low-resource workflows.

Chapter 2: Literature Review and Gap Analysis

2.1 Health Informatics in Low-Resource Setting Workflows

Health informatics in low-resource settings must account for workflow realities that differ from well-connected hospital environments. Rural facilities may depend on paper registers, batch reporting, shared devices, intermittent electricity, and limited technical support. Cloud-only tools can fail when continuous connectivity is assumed; therefore, an offline-first design is not an optional enhancement but a core resilience requirement.

Digital health literature emphasizes that technology must fit the health-system context. For AfyaIntel Hub, this means local data capture must be possible without internet access, synchronization must tolerate failure and retry safely, and the system must preserve clinical accountability even when AI assistance is available.

2.2 Clinical Natural Language Processing for African Languages

Clinical NLP systems in high-resource languages benefit from large annotated corpora, mature medical vocabularies, and domain-specific models. Swahili and other African languages remain comparatively under-resourced, especially in clinical domains. Existing models such as SwahBERT, multilingual BERT fine-tuned for Swahili, and AfroXLMR provide a practical starting point but require clinical adaptation, local vocabulary expansion, and rigorous safety evaluation before use in health workflows.

AfyaIntel Hub therefore treats local-language NLP as a supervised, clinically reviewed component. The proposed NLP pipeline uses normalization of Swahili and Swanglish expressions, dictionary-backed synonym handling, red-flag keyword detection, and transformer-based classification. The model output is not a diagnosis; it is a structured cue for human review.

2.3 Mathematical Forecasting in Pharmaceutical Supply Chains

Medicine supply forecasting in low-resource settings is complicated by irregular deliveries, incomplete ledgers, emergency redistribution, seasonal disease patterns, and stock movements that may not be recorded in real time. Traditional methods such as moving averages and exponential smoothing are easy to interpret but may perform poorly when nonlinear patterns and missing data dominate. Machine learning methods such as Random Forest and XGBoost can model nonlinear relationships using lagged consumption, seasonality, facility type, recent diagnosis volumes, and stock movement features. Prophet provides interpretable seasonality and holiday effects and can serve as a baseline or complementary forecasting model.

2.4 Interoperability Frameworks in Public Health

Interoperability requires more than API connectivity. It also requires semantic agreement about what a patient, encounter, observation, medicine order, facility, stock movement, and aggregate indicator mean. HL7 FHIR R4 provides a resource-based standard for representing clinical and administrative data in structured JSON or XML formats. In Tanzania, AfyaIntel Hub should not attempt to replace national platforms such as GoT-HOMIS or DHIS2. Instead, it should function as a safe mapping and synchronization layer that can export approved clinical or aggregate data in formats aligned with those systems.

2.5 Research Gap Identification

Table 4. Research Gap Matrix

Existing approach	Limitation	AfyaIntel Hub response
General medical chatbots	Often require stable connectivity, may not support local clinical governance, and may not map to supply chain needs.	Offline-first, human-reviewed, local-language triage support connected to inventory and forecasting.
Static spreadsheets	Useful for local recordkeeping but weak for NLP, synchronization, audit trails, and predictive alerts.	Structured database, encrypted sync queue, forecast engine, dashboards, and role-based access.
Standalone EMR systems	Focus primarily on facility records and may not provide localized NLP or stock-out forecasting.	Interoperable intelligence layer that complements official EMRs rather than replacing them.
National aggregate reporting platforms	Effective for public health reporting but may depend on delayed facility submissions.	Point-of-care signal capture and controlled aggregate mapping to DHIS2-style indicators.
Supply logistics platforms	May focus on stock balances without symptom trends or clinical workload indicators.	Disease-aware supply forecasting using clinical and inventory features under governance controls.

Chapter 3: Research Methodology (DSRM Framework)

3.1 DSRM Research Design

This study applies Design Science Research Methodology because the central contribution is a purposeful software artifact designed to solve a practical health-service problem while producing transferable design knowledge. DSRM is appropriate where the research objective is not only to observe a phenomenon but to design, build, demonstrate, evaluate, and communicate an artifact that changes practice.

The AfyaIntel Hub artifact is evaluated according to its utility, technical feasibility, workflow fit, interoperability readiness, and safety governance. The methodology follows the sequence of problem identification, objectives of a solution, design and development, demonstration, evaluation, and communication.

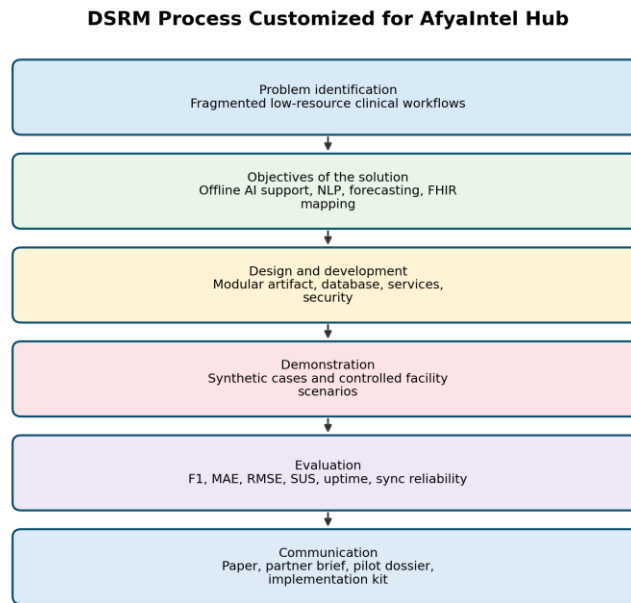


Figure 1. Design Science Research Methodology process customized for AfyaIntel Hub.

3.2 Iterative Development Framework

Table 5. Iterative Development Roadmap

Phase	Software maturity	Primary output
Phase 1	Standalone prototype	SQLite-based offline tablet module for symptom entry, vitals capture, and medicine stock lookup.
Phase 2	Modular local services	NLP preprocessing service, local clinical dictionary, role-based login, and encrypted queue.
Phase 3	Containerized API architecture	Central PostgreSQL service, FHIR mapper, dashboard service, model server, and sync API.
Phase 4	Controlled pilot readiness	Synthetic test pack, clinical QA checklist, facility readiness assessment, and data protection checklist.
Phase 5	Institutional pilot	Ethically approved deployment in selected facilities with real-

Phase	Software maturity	Primary output
		world workflow evaluation.

3.3 Health Data Requirements

Table 6. Health Data Requirement Matrix

Data category	Sample fields	Intended use
Patient encounter	Encounter ID, facility code, visit date, age group, sex, residence area, clinician ID.	Link symptom entries to point-of-care service events while minimizing identifiable fields.
Symptom note	Raw Swahili/Swanglish text, normalized text, extracted symptoms, urgency tags, red flags.	NLP parsing, triage support, surveillance aggregation, and quality review.
Clinical observation	Temperature, weight, blood pressure, pregnancy status where applicable, RDT result, lab flag.	Triage support, feature inputs for disease trends, and clinical prompt filtering.
Medicine inventory	Item code, batch optional, stock balance, daily issues, receipts, losses, expiry date, reorder level.	Forecasting, stock-out alerts, inventory dashboard, and supply prioritization.
Facility readiness	Connectivity status, power source, device count, staffing, stock-management maturity.	Deployment planning and interpretation of performance outcomes.
System audit	User ID, role, timestamp, action, device ID, sync status, model version.	Security accountability, clinical governance, and data protection compliance.

3.4 Clinical Facility Readiness Levels

Table 7. Clinical Facility Readiness Classification

Level	Readiness profile	Recommended deployment mode
Level 1: Paper-only	Patient and stock records are maintained in paper registers; no routine digital data entry at the point of care.	Standalone tablet, daily local backup, supervised digitization, and offline-first workflows.
Level 2: Hybrid	Paper records are primary, but selected reports are entered into	Tablet-based capture, local SQLite cache, periodic

Level	Readiness profile	Recommended deployment mode
	spreadsheets or district systems later.	synchronization, and dashboard summaries.
Level 3: Partially digitized	Facility has local computers or LAN; selected clinical or stock workflows are computerized.	Local network deployment, API-based synchronization, FHIR mapping, and role-based dashboards.
Level 4: Connected facility	Facility has stable connectivity and participates in routine digital reporting.	Real-time or near-real-time synchronization with central server and external systems.

3.5 Evaluation Strategy

The evaluation strategy is designed to avoid unsupported claims. Field performance must be measured after ethical clearance and pilot authorization. Before real patient data are used, the artifact should be evaluated with synthetic cases, de-identified historical aggregates where authorized, and clinician-reviewed simulation scenarios.

Table 8. Evaluation Metrics

Evaluation dimension	Metric	Purpose
NLP extraction	Precision, recall, F1-score, confusion matrix	Quantify extraction quality for symptoms, red flags, and intent categories.
Forecasting	MAE, RMSE, WAPE, stock-out lead-time accuracy	Measure predicted medicine consumption and depletion-window accuracy.
Offline reliability	Sync success rate, average queue delay, conflict resolution accuracy	Assess whether work continues safely during connectivity loss.
Usability	System Usability Scale (SUS), task completion time, error frequency	Assess workflow fit among rural clinical officers and facility storekeepers.
Clinical safety	Human review completion rate, AI override rate, red-flag escalation accuracy	Ensure AI outputs remain under human clinical control.
Interoperability	FHIR validation pass rate, field completeness, mapping coverage	Assess readiness for data exchange with GoT-HOMIS and DHIS2-style workflows.

Chapter 4: Conceptual Framework and System Design

4.1 Conceptual Framework Diagram

The conceptual framework begins at the point of care, where a clinician or community health worker records a patient's symptom narrative in Swahili, English, or Swanglish. The offline NLP gateway normalizes the text, extracts clinical intents and red flags, and produces structured support prompts. The system then checks local medicine stock, estimates near-term supply risk, stores the event locally, and synchronizes approved records to a regional dashboard when connectivity returns.

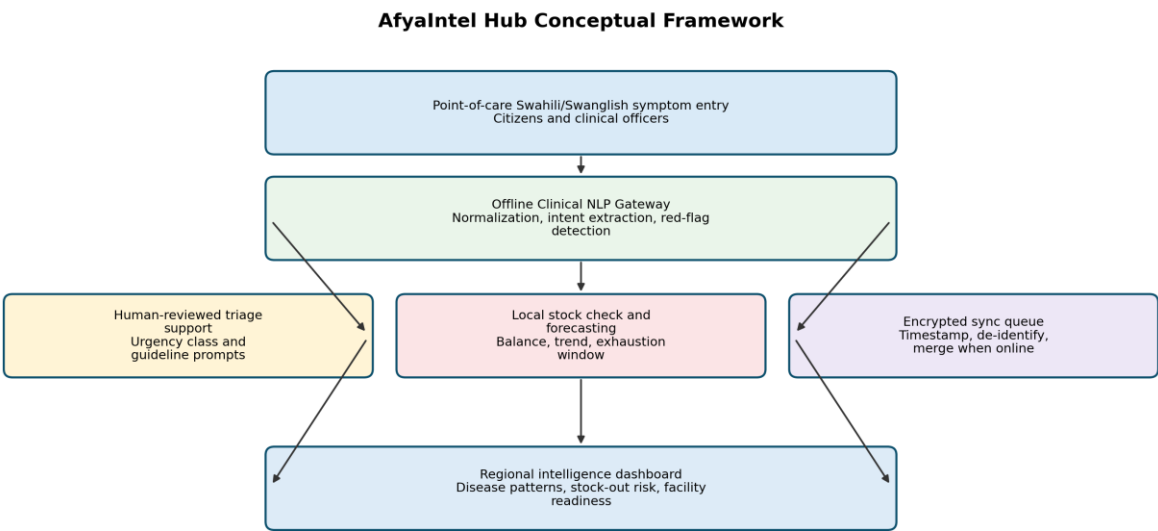


Figure 2. Conceptual framework showing point-of-care symptom entry, offline NLP, stock checks, synchronization, and regional analytics.

4.2 System Architecture

AfyaIntel Hub uses a multi-tier architecture composed of a presentation layer, application services, local and central data stores, AI/ML services, security controls, interoperability services, and governance workflows. The design supports progressive deployment, meaning a facility can begin with offline local functionality and later integrate with central dashboards and national systems as infrastructure matures.

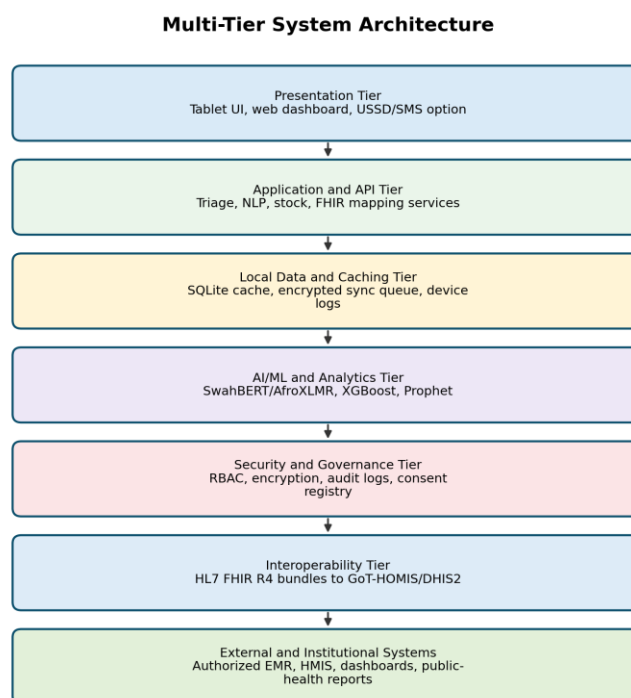


Figure 3. Multi-tier system architecture for offline-first health intelligence and interoperability.

4.3 System Modules

Table 9. Core System Modules

Module	Function	Key technical features
Offline Triage Gateway	Captures patient symptom descriptions, vitals, and initial urgency indicators without active internet.	Tablet UI, SQLite cache, local validation, red-flag vocabulary, QR/printed summary option.
Swahili Clinical NLP Pipeline	Normalizes informal Swahili/Swanglish notes and structures them into medical tags.	Tokenizer, spelling normalization, dictionary expansion, SwahBERT/AfroXLMR classifier, intent and entity extractor.
Supply Chain Forecasting Engine	Evaluates stock balances, consumption patterns, and likely exhaustion windows.	XGBoost for nonlinear features, Prophet for seasonality baseline, missing-data imputation, reorder alerts.
FHIR Interoperability Mapper	Converts local database records into standard resources for approved exchange.	FHIR R4 Patient, Encounter, Observation, MedicationRequest,

Module	Function	Key technical features
		Organization, Location, and aggregate indicator mapping.
Regional Analytics Dashboard	Displays aggregate disease signals, facility readiness, medicine risk, and data quality indicators.	Role-filtered dashboards, geospatial summaries, stock-out risk ranking, audit review panels.

4.4 Data Flow and Synchronization Controls

AfyaIntel Hub records local transactions with a unique device identifier, facility code, event timestamp, user role, and sync status. Records are validated locally, encrypted, and stored in a queue. When connectivity is detected, the queue uploads records to a central server that verifies schema, checks for duplicates, resolves conflicts, updates dashboards, and returns an acknowledgement to the device.

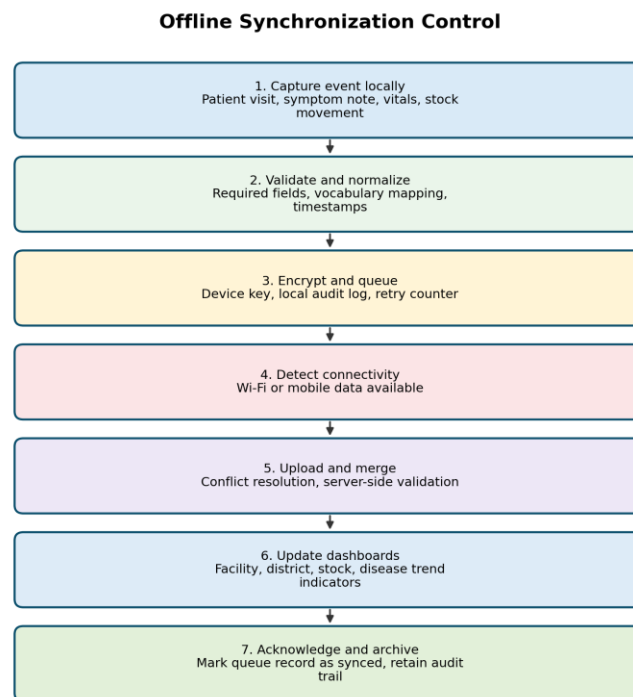


Figure 4. Offline data flow and synchronization control sequence.

Chapter 5: Database Design, AI/ML Workflow, and Mathematical Framework

5.1 Logical Database Design

Table 10. Logical Database Schema

Table	Primary fields	Purpose
patients	patient_uuid, sex, age_years, village_code, consent_status, created_at	Stores minimized patient profile data or pseudonymous identifiers.
encounters	encounter_uuid, patient_uuid, facility_id, clinician_id, visit_datetime, encounter_type	Represents a point-of-care service event.
symptom_notes	note_id, encounter_uuid, raw_text, normalized_text, language_tag, model_version	Stores clinical narratives and NLP processing status.
nlp_entities	entity_id, note_id, entity_type, entity_text, mapped_code, confidence_score	Stores extracted symptoms, signs, intents, and mapped codes.
clinical_observations	observation_id, encounter_uuid, observation_type, value, unit, abnormal_flag	Stores vitals, RDT results, and selected lab flags.
drug_inventory	inventory_id, facility_id, drug_code, stock_balance, reorder_level, expiry_date, updated_at	Tracks local medicine availability.
stock_movements	movement_id, facility_id, drug_code, movement_type, quantity, movement_date	Tracks receipts, issues, adjustments, expiries, and losses.
forecasts	forecast_id, facility_id, drug_code, horizon_days, predicted_consumption, stockout_risk	Stores generated medicine demand and risk forecasts.
sync_queue	queue_id, entity_type, entity_uuid, payload_hash, status, retry_count	Manages offline-to-online synchronization.
system_logs	log_id, user_id, role, action, timestamp, device_id, ip_address	Supports auditability and security review.

5.2 Clinical AI/ML Workflow Sequence

The clinical AI workflow begins with raw text capture and ends with human-reviewed structured support. The system first detects language patterns, normalizes spelling and common abbreviations, identifies candidate medical concepts, passes the normalized text into a transformer-based classifier, produces structured urgency and symptom labels, and stores outputs for clinician review.

Raw clinical note -> language detection -> spelling and dialect normalization -> symptom dictionary lookup -> SwahBERT/AfroXLMR inference -> entity and intent extraction -> red-flag rule check -> clinician review -> confirmed structured record -> audit log and optional FHIR export

5.3 Mathematical Models

5.3.1 Clinical Notation and Structural Variables

Table 11. Variable Notation

Symbol	Meaning
p	Unique patient or pseudonymous patient profile.
l	Clinical location or facility.
m	Medical condition, symptom group, or diagnostic category.
s	Pharmaceutical supply item or medicine code.
t	Time index, usually day or week.
h	Forecast horizon.
x_t	Feature vector at time t, including stock balance, consumption, seasonality, and diagnosis volume.
y_{s,l,t+h}	Predicted consumption or stock need for medicine s at location l over horizon h.

5.3.2 Swahili Medical NLP Triage Parsing Logic

Let a raw clinical note be represented as N. The preprocessing function g(N) performs lowercasing, spelling normalization, Swahili/English code-mixed token handling, and removal of non-clinical noise. The transformer encoder E maps the cleaned note into contextual embeddings. The classifier C produces a probability distribution over medical intent labels such as urgent referral, non-urgent review, health education, medicine availability check, or red-flag escalation.

```
N_clean = g(N)
H = E(N_clean)
y_intent = argmax C(H)
entities = EntityExtractor(H, clinical_dictionary, red_flag_rules)
```

```
final_output = clinician_review(y_intent, entities,
confidence_scores)
```

The mapping to standardized codes should proceed conservatively. Informal expressions are mapped to symptom and condition dictionaries, then reviewed against ICD-10 or locally approved clinical terminologies. No automated mapping should be written as a final diagnosis without clinician confirmation.

5.3.3 Drug Supply Consumption Forecasting Model

For each medicine s at facility l , the forecasting model estimates demand over a future horizon h using historical consumption, stock receipts, losses, seasonal indicators, facility workload, recent syndrome counts, and known stock-out interruptions. XGBoost is selected for its ability to handle nonlinear relationships and mixed feature sets, while Prophet is used as an interpretable time-series baseline for trend and seasonality.

```
Input features  $x_{\{s,l,t\}} = [\text{consumption\_lags}, \text{stock\_balance},$ 
 $\text{receipts}, \text{losses}, \text{month}, \text{rainfall\_proxy}, \text{malaria\_like\_cases},$ 
 $\text{facility\_level}, \text{stockout\_history}]$ 
XGBoost objective: minimize  $\sum L(y_{\{s,l,t+h\}}, f(x_{\{s,l,t\}})) + \Omega(f)$ 
Prophet baseline:  $y(t) = \text{trend}(t) + \text{seasonality}(t) +$ 
 $\text{holiday\_or\_event\_effect}(t) + \text{error}_t$ 
Stock-out risk = 1 if  $\text{predicted\_available\_stock}_{\{t+h\}} <$ 
 $\text{reorder\_threshold}_s$ 
```

5.3.4 Outbreak and Epidemiological Risk Scoring

The epidemiological risk score combines sudden increases in symptom clusters, abnormal diagnostic volumes, facility workload, and local stock pressure. The model should be interpreted as a surveillance cue rather than an outbreak declaration.

```
RiskScore_{m,l,t} = w1 * z(symptom_count_{m,l,t}) + w2 *
z(positive_lab_flags_{m,l,t}) + w3 * z(visit_volume_{l,t}) + w4 *
stock_pressure_{s,l,t}
Alert if RiskScore_{m,l,t} >= threshold_m and clinician/public-health
review confirms plausibility
```

5.4 Performance Evaluation Metrics

Table 12. Mathematical Evaluation Metrics

Metric	Formula	Interpretation
Precision	$TP / (TP + FP)$	Proportion of extracted labels that were correct.
Recall	$TP / (TP + FN)$	Proportion of true labels successfully extracted.
F1-score	$2 * \text{Precision} * \text{Recall} /$ $(\text{Precision} + \text{Recall})$	Balanced score for NLP extraction quality.
MAE	$\text{mean}(y_t - \hat{y}_t)$	Average absolute forecasting

Metric	Formula	Interpretation
		error.
RMSE	$\sqrt{\text{mean}((y_t - \hat{y}_t)^2)}$	Forecasting error that penalizes large deviations.
WAPE	$\text{sum}(y_t - \hat{y}_t) / \text{sum}(y_t)$	Scale-normalized absolute forecasting error.
SUS	Standard 10-item System Usability Scale	Usability score for clinical officer and storekeeper workflows.

Chapter 6: Multi-Layer Operational Framework and Cost Model

6.1 Capital and Infrastructure Categories

Table 13. Capital and Infrastructure Categories

Category	Items	Planning rationale
Clinical devices	Rugged tablets, protective cases, barcode/QR support where feasible.	Supports point-of-care use in facilities where desktop workstations are impractical.
Power resilience	Solar charging kits, power banks, surge protectors.	Reduces downtime caused by unreliable electricity.
Local networking	Local router, secure Wi-Fi, optional LAN storage.	Supports facility-level synchronization between devices even before internet is available.
Server infrastructure	Central cloud or regional server, PostgreSQL, backup storage, monitoring.	Hosts dashboards, APIs, model services, and audit repositories.
Security infrastructure	Device management, encryption keys, backup media, access-control administration.	Protects clinical and operational data.

6.2 Operational Unit Costs

Table 14. Operational Unit Cost Framework

Cost driver	Unit basis	Budget logic
Training	Per clinical worker, storekeeper, and health information officer.	Covers workflow training, clinical safety rules, data protection, and troubleshooting.
Hosting and backup	Monthly central infrastructure cost.	Covers server operation, database backup, secure logs, and monitoring.
Software maintenance	Monthly or quarterly development cycle.	Covers bug fixes, model updates, security patches, and dashboard improvements.
Clinical QA	Review sessions and expert validation panels.	Ensures vocabulary, red flags, and triage prompts remain clinically safe.
Field support	Facility visits, remote support, and device maintenance.	Keeps rural deployments operational.

6.3 Multi-Phase Deployment Horizon

Table 15. Deployment Horizon and Milestone Budget Map

Horizon	Deployment scale	Milestone evidence
0-3 months	Laboratory prototype and synthetic test dataset	Working offline app, sample dashboards, NLP test pack, and data dictionary.
4-6 months	Single clinic demonstration	Usability feedback, sync logs, stock lookup tests, and clinician review checklist.
7-12 months	Controlled pilot in 3-5 facilities	F1-score report, forecasting error report, SUS score, and interoperability validation.
13-24 months	District-level pilot	Facility comparisons, disease/stock dashboards, data quality audit, and governance report.
24+ months	Regional scale-up preparation	Implementation toolkit, sustainability model, partner agreements, and regulatory dossier.

6.4 Sustainability Strategy

AfyIntel Hub should be sustained through a blended model combining public health partnerships, research grants, responsible donor support, district-level implementation agreements, and service contracts for training, maintenance, hosting, and interoperability support. The system should avoid dependence on a single donor by developing modular deployment packages that can be adopted progressively by facilities with different readiness levels.

Chapter 7: Security, Privacy, and Human-in-the-Loop Governance

7.1 Privacy Compliance Matrix

Table 16. Privacy Compliance Matrix

Governance requirement	AfyIntel Hub control	Evidence artifact
Lawful and fair processing	Collect data only under approved clinical workflow, facility authorization, and consent or legal basis.	Data protection impact assessment and consent/authorization records.

Governance requirement	AfyaIntel Hub control	Evidence artifact
Purpose limitation	Separate clinical care, research analytics, and aggregate reporting purposes.	Role-based purpose registry and data-use policy.
Data minimization	Use pseudonymous IDs and avoid unnecessary personal identifiers in analytics.	Schema review and minimized field dictionary.
Security safeguards	Encrypt local databases, secure APIs, use RBAC, and retain audit logs.	Security architecture document and penetration test report.
Accountability	Assign data controller/processor roles and maintain processing logs.	Governance register and audit trail.
Data subject rights	Define procedures for correction, withdrawal where applicable, and authorized access requests.	Standard operating procedure for data requests.

7.2 Security Architecture

The security architecture uses defense in depth. The local application encrypts stored records, restricts user access by role, logs all sensitive actions, signs synchronization payloads, and uses secure transport when exchanging data with the server. The central system separates identifiers from analytics where possible and applies access rules based on user role, facility, and approved data purpose.

Table 17. Role-Based Access Control Matrix

Role	Permitted functions	Restricted functions
Clinical officer or nurse	Enter encounters, review AI prompts, confirm clinical actions, view local patient summaries.	Cannot export raw identifiable datasets or change model rules.
Facility storekeeper/pharmacist	Manage stock balances, receipts, issues, expiry alerts, and stock forecasts.	Cannot access detailed symptom narratives unless authorized for clinical workflow.
Health information officer	Review data quality, sync status, aggregate reports, and facility dashboards.	Cannot override clinical decisions or prescribe.
District manager	View aggregate facility indicators, stock risk, disease trends, and data completeness.	Cannot view unnecessary patient identifiers.

Role	Permitted functions	Restricted functions
System administrator	Manage users, roles, backups, device registration, and logs.	Cannot alter clinical records without audit and authorization.

7.3 Human-in-the-Loop Clinical Control

Human-in-the-loop governance is a central safety requirement. AfyaIntel Hub may classify urgency, extract red flags, propose guideline references, or highlight stock issues, but a qualified human user must review and record the final clinical decision. The system should display confidence scores and uncertainty warnings rather than hiding model limitations.

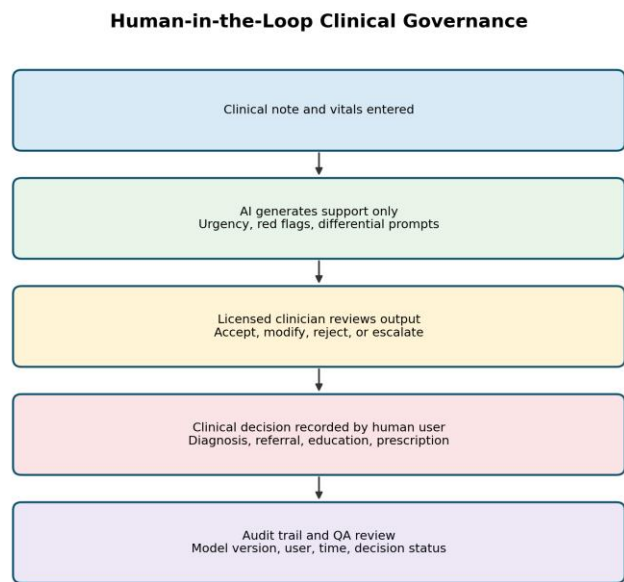


Figure 5. Human-in-the-loop clinical governance workflow.

Chapter 8: Prototyping, Pilot Evaluation, and System Performance

8.1 Prototype Scope (MVP)

Table 18. MVP Scope Boundary

Included in MVP	Excluded from MVP
Offline Swahili/Swanglish symptom logging	Autonomous diagnosis or automated prescription.
Basic vitals and red-flag prompts	Replacement of national EMR or regulatory clinical systems.
Local drug inventory lookup	Full national MSD integration without formal partnership.
Synthetic and de-identified demonstration datasets	Unapproved use of identifiable patient records.
FHIR-compatible export prototype	Live national data exchange before authorization.

8.2 Structured Pilot Dataset Layout

Table 19. Pilot Dataset Layout

Dataset	Fields	Privacy treatment
Synthetic patient encounters	Age band, sex, symptom text, vitals, urgency label, clinician-reviewed label.	No real patient identifiers.
Annotated NLP corpus	Raw Swahili/Swanglish note, normalized note, symptom entities, red flags, intent label.	De-identified or synthetic; clinical reviewer approval required.
Medicine stock history	Drug code, opening balance, receipts, issues, losses, closing balance, stock-out days.	Facility-coded; no patient identifiers.
Usability logs	Task time, error count, SUS response, user role, facility readiness level.	User consent and pseudonymous staff IDs.
Sync performance logs	Payload size, retry count, queue delay, merge result, conflict status.	System metadata only.

8.3 Demonstration Scenarios

Table 20. Demonstration Scenario Workflow

Scenario	Workflow	Measured output
----------	----------	-----------------

Scenario	Workflow	Measured output
Swahili symptom entry	Clinical worker enters informal Swahili note; NLP extracts symptoms and red flags; clinician confirms or corrects.	F1-score by entity type and clinician correction rate.
Offline stock check	Facility is offline; storekeeper records stock issue; system calculates remaining days of stock.	Local response time and forecast calculation success.
Delayed synchronization	Device stores 100 events offline; connectivity returns; encrypted queue uploads records.	Sync success rate, average queue delay, conflict rate.
Aggregate dashboard update	Approved data are merged; district dashboard updates disease and stock indicators.	Dashboard refresh time and field completeness.
Human override	AI flags high urgency; clinician reviews and either confirms or overrides with reason.	Override rate and audit completeness.

8.4 System Performance Outcomes

System performance should be reported after controlled implementation. The following table defines the final results structure to be used after a pilot. No real-world clinical performance claim should be inserted until the artifact is tested with approved data and reviewed by qualified clinical and data-governance teams.

Table 21. System Performance Reporting Template

Metric	Target reporting value	Evidence source
NLP symptom extraction F1-score	To be populated after annotated corpus evaluation.	Clinician-annotated Swahili/Swanglish corpus.
Forecast MAE/RMSE/WAPE	To be populated after historical or simulated stock-data testing.	Medicine stock movement dataset.
Local database load time	To be populated after tablet performance benchmarking.	Device logs under defined payload sizes.
Synchronization success rate	To be populated after network outage simulation.	Sync queue logs and server acknowledgements.
Tablet battery depletion rate	To be populated after controlled runtime tests.	Device telemetry during clinical workflow simulation.

Metric	Target reporting value	Evidence source
System Usability Scale score	To be populated after clinical officer and storekeeper testing.	SUS questionnaire and task observations.

Chapter 9: Scientific Contribution and Practical Horizons

9.1 Informatics Contributions

AfyaIntel Hub contributes a localized design model for connecting clinical NLP, offline-first data capture, predictive medicine supply planning, and interoperability in low-resource primary care. The contribution is not a single algorithm but an integrated health informatics artifact that treats language, infrastructure, governance, and clinical workflow as interdependent design requirements.

9.2 Practical Impact

If validated through ethical pilots, AfyaIntel Hub may improve structured symptom documentation, reduce delays in recognizing stock risks, support more timely aggregate disease intelligence, and make clinical decision-support tools more usable in rural facilities with intermittent connectivity. The strongest practical value is workflow resilience: the system remains useful when internet access is absent and synchronizes when connectivity returns.

9.3 Research Limitations

Table 22. Research Limitations and Mitigation

Limitation	Risk	Mitigation strategy
Limited clinical training corpus	NLP model may underperform on rare phrases and dialects.	Use clinician-reviewed annotation, active learning, and regional vocabulary expansion.
Dialect and code-mixing variation	Different regions may express symptoms differently.	Evaluate by dialect subset and include local health workers in vocabulary review.
Incomplete stock histories	Forecasting error may be high where historical records are irregular.	Use missing-data indicators, robust baselines, and storekeeper validation.
Device durability	Tablets may fail under harsh facility conditions.	Use protective casing, power backups, replacement plans, and offline backup procedures.
Governance complexity	Data sharing may be delayed by regulatory and institutional approvals.	Prepare data protection assessment, ethical protocol, and partner memoranda before pilot.

9.4 Recommendations and Conclusion

AfyaIntel Hub should proceed through a controlled pathway: technical prototype validation, clinical vocabulary review, synthetic dataset testing, facility readiness assessment, ethical clearance, limited pilot deployment, and staged scale-up. National scaling should occur only through formal partnerships with health authorities, clinical institutions, data protection bodies, and interoperability stakeholders.

The future of African digital health infrastructure depends on systems that are locally grounded, language-aware, resilient to infrastructure constraints, interoperable with national systems, and

accountable to human clinical judgment. AfyaIntel Hub provides a Design Science Research contribution toward that future by demonstrating how a generative health intelligence framework can support low-resource care without erasing clinical authority or data sovereignty.

References

- Alabi, J. O., Adelani, D. I., Mosbach, M., & Klakow, D. (2022). Adapting pre-trained language models to African languages via multilingual adaptive fine-tuning. *Proceedings of COLING 2022*.
- Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*.
- HL7 International. (2019). HL7 FHIR Release 4 specification. Health Level Seven International.
- Ministry of Health, Community Development, Gender, Elderly and Children. (2019). *The National Digital Health Strategy 2019-2024*. United Republic of Tanzania.
- Ministry of Health, Community Development, Gender, Elderly and Children. (2021). *Standard Treatment Guidelines and National Essential Medicines List for Tanzania Mainland, Sixth Edition*. United Republic of Tanzania.
- Peppers, K., Tuunanen, T., Rothenberger, M. A., & Chatterjee, S. (2007). A design science research methodology for information systems research. *Journal of Management Information Systems*, 24(3), 45-77.
- Taylor, S. J., & Letham, B. (2018). Forecasting at scale. *The American Statistician*, 72(1), 37-45.
- United Republic of Tanzania. (2022). *The Personal Data Protection Act, 2022*.
- World Health Organization. (2019). *WHO guideline: recommendations on digital interventions for health system strengthening*. World Health Organization.
- DHIS2. (n.d.). *DHIS2 health management information system documentation and implementation resources*. HISP Centre, University of Oslo.
- Hugging Face. (n.d.). *SwahBERT and multilingual BERT models fine-tuned for Swahili language tasks*. Hugging Face Model Hub.

Appendix A: Sample Generated System Interfaces

The following interface descriptions are provided as text-based wireframes to support implementation without relying on non-editable screenshots.

Clinical Tablet Screen

Facility: [Dispensary Name] Mode: OFFLINE
Patient ID: [Pseudonymous ID] Age: [] Sex: []
Symptom note: [Ana homa, kikohozi, maumivu ya kichwa...]
Vitals: Temp [] Weight [] BP [] RDT []
[Run NLP Support] [Check Stock] [Save Encounter]
AI support: Red flags [] Urgency [] Suggested guideline reference []
Clinician decision: [Confirm] [Modify] [Reject] [Refer]

Central Inventory Dashboard

Region/District: [] Reporting week: []

Medicine	Stock days left	Forecast risk	Action
ALu	9	High	Reorder
Amoxicillin	18	Medium	Monitor
ORS	35	Low	Normal

Sync status: 92% facilities reporting; 5 devices offline

Appendix B: Complete System Data Dictionary

Table B1. Data Dictionary

Field	Type/length	Constraints	Description
patient_uuid	UUID	Primary key; pseudonymous	Unique patient identifier generated locally.
facility_id	VARCHAR(50)	Foreign key	Facility identifier aligned with approved facility registry.
raw_text	TEXT	Encrypted at rest	Original symptom narrative captured at the point of care.
normalized_text	TEXT	Derived field	Cleaned and normalized clinical text used for NLP.
language_tag	VARCHAR(20)	Allowed: sw, en, sw-en	Detected language or code-mixed label.
confidence_score	DECIMAL(5,4)	0.0000-1.0000	Model confidence for extracted label.
drug_code	VARCHAR(50)	Mapped to NEMLIT/local item code	Medicine identifier used in inventory and forecasts.
stock_balance	INTEGER	≥ 0	Current facility-level quantity on hand.
sync_status	VARCHAR(20)	queued, sent, confirmed, failed	Synchronization state of local record.
audit_action	VARCHAR(100)	Required	Action performed by authenticated user.

Appendix C: Clinical Quality Assurance Checklist

Table C1. Clinical Quality Assurance Checklist

Checklist item	Verification rule	Responsible reviewer
Medical vocabulary safety	All symptom, red-flag, and medication terms reviewed against approved clinical references.	Clinical reviewer
AI output boundary	System labels output as support only, not final diagnosis.	Clinical governance team
Dosage and medicine prompts	Any dosage support requires guideline source and human confirmation.	Pharmacist/clinician
Data protection	Only necessary data fields are collected and role-based access is enforced.	Data protection lead
Interoperability validation	FHIR exports pass schema validation before external exchange.	Health informatics reviewer
Pilot readiness	Facility readiness, training completion, backup plan, and incident reporting are confirmed.	Pilot coordinator

Appendix D: FHIR Mapping to GoT-HOMIS and DHIS2-Oriented Outputs

Table D1. FHIR Mapping Layer

Local AfyaIntel field	FHIR R4 resource	FHIR element	Target use
patient_uuid, sex, age	Patient	Patient.identifier, Patient.gender, Patient.birthDate or age extension	GoT-HOMIS-compatible patient or pseudonymous record where authorized.
encounter_uuid, visit_datetime, facility_id	Encounter	Encounter.identifier, Encounter.period, Encounter.serviceProvider	Facility visit event.
temperature, RDT result, lab flag	Observation	Observation.code, Observation.value[x], Observation.effectiveDateTime	Clinical observation and aggregate

Local AfyaIntel field	FHIR R4 resource	FHIR element	Target use
			indicator source.
symptom_entities	Condition or Observation	Condition.code or Observation.code	Structured symptom or condition candidate after clinical review.
drug_code, prescribed_item	MedicationRequest	MedicationRequest.medicationCodeableConcept	Clinician-confirmed medicine request where permitted.
facility stock balance	InventoryReport or Observation	InventoryReport.inventoryListing.item.item, quantity	Facility-level stock reporting or local supply dashboard.
aggregate malaria-like cases	MeasureReport or DHIS2 data element	MeasureReport.group.measureScore	DHIS2-oriented aggregate disease indicator after approval.